Laboratory Tasks

Task 1 : Leap Year Check

```c
#include <stdio.h>

int main () {
    int n;
    printf("enter a year: ");
    scanf("%d", &n);
    if (n % 100 != 0 && n % 4 ==0) {
        printf("the year is leap\n");
    }

    else if (n % 4 == 0 && n % 100 == 0 && n % 400 == 0) {
        printf("the year is leap\n");
    }

    else printf("the year is not leap\n");
    return 0;
}
```

```
enter a year: 2000
the year is leap


...Program finished with exit code 0
Press ENTER to exit console.
```

Task 2 : Vowel or Consonant

```c
#include <stdio.h>

int main () {
    char c;
    printf("enter a letter: ");
    scanf("%c", &c);

    switch (c) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
        printf("the letter you have entered is a vowel");
        break;

    default:
    printf("the letter you have entered is not a vowel");

    }

    return 0;
}
```

```
enter a letter: r
the letter you have entered is not a vowel

...Program finished with exit code 0
Press ENTER to exit console.
```

Task 3 : Menu-Driven Calculator

```c
#include <stdio.h>

int main () {
    char c;
    float a, b;

    printf("you have 5 operations available: +, -, *, /, %\n");
    printf("enter 2 numbers: ");
    scanf("%f%f", &a, &b);
    printf("enter the operation: ");
    scanf(" %c", &c);

    switch (c) {
        case '+':
        printf("result: %.2f\n", a + b);
        break;

        case '-':
        printf("result: %.2f\n", a - b);
        break;

        case '*':
        printf("result: %.2f\n", a * b);
        break;

        case '/':
        printf("result: %.2f\n", a / b);
        break;

        case '%':
        printf("result: %.f\n", (int)a % (int)b);
        break;

    default:
    printf("invalid");

    }

    return 0;

}
```

```
you have 5 operations available: +, -, *, /, %
enter 2 numbers: 4
6
enter the operation: +
result: 10.00


...Program finished with exit code 0
Press ENTER to exit console.
```

Additional Practice Tasks

Write a C program to check whether a given number is positive, negative, or zero.

```c
#include <stdio.h>

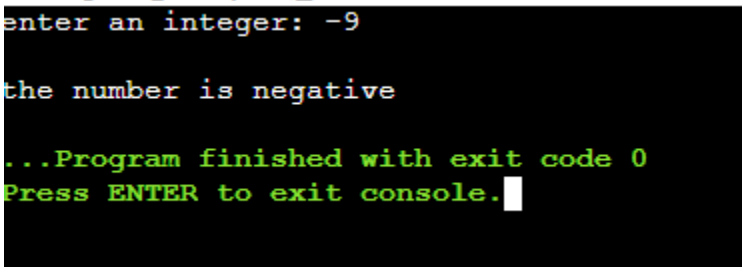
int main () {
    int n;
    printf("enter an integer: ");
    scanf("%d", &n);

    if (n == 0)
    printf("\nthe number is zero");

    else if (n < 0)
    printf ("\nthe number is negative");

    else printf("\nthe number is positive");

    return 0;
}
```



Write a C program to find the largest of three numbers using nested if-else.

```c
#include <stdio.h>

int main () {
    int a, b, c;
    printf("enter 3 numbers: ");
    scanf("%d%d%d", &a, &b, &c);

    if (a > b) {
        if (a > c) {
            printf("%d is the greatest number\n", a);

        } else {
            printf("%d is the greatest number\n", c);
        }

    } else {
        if (b > c) {
            printf("%d is the greatest number\n", b);

        } else {
            printf("%d is the greatest number\n", c);
        }
    }

    return 0;
}
```

```
enter 3 numbers: 546
436
437
546 is the greatest number


...Program finished with exit code 0
Press ENTER to exit console.
```

Write a C program that accepts the three angles of a triangle and first checks whether they form a valid triangle (the angles must sum to 180◦); if valid, classify it as Equilateral, Isosceles, or Scalene.

```c
#include <stdio.h>

int main () {
    int a, b, c;
    printf("enter 3 angles of a triangle: ");
    scanf("%d%d%d", &a, &b, &c);

    if (a + b + c == 180) {
        printf("this is a valid triangle\n");

        if (a == b == c) {
            printf("it is an equilateral triangle\n");
        }

        else if (a == b || b == c || a == c) {
            printf("it is an isoceles triangle\n");
        }

        else {
            printf("it is a scalene triangle\n");
        }
    }

      else {
         printf("this is not a valid triangle\n");
    }

return 0;

}
```

```
enter 3 angles of a triangle: 70
50
60
this is a valid triangle
it is a scalene triangle


...Program finished with exit code 0
Press ENTER to exit console.
```

Write a C program to validate a 4-digit PIN entered by the user against a stored constant PIN, and print "Access Granted" or "Access Denied".

```c
#include <stdio.h>
#define pincode 6767

int main () {
    int n;
    printf("enter pin:\n");
    scanf("%d", &n);

    if (n == pincode) {
        printf("access granted\n");
    } else {
        printf("access denied\n");
    }

    return 0;
}
```

```
enter pin:
6769
access denied


...Program finished with exit code 0
Press ENTER to exit console.
```

Write a C program that accepts a day number (1–7) and prints the corresponding day name using switch-case.

```c
#include <stdio.h>

int main () {
    int n;
    printf("enter a number from 1-7: ");
    scanf("%d", &n);

    switch (n) {
        case 1:
        printf("monday");
        break;

        case 2:
        printf("tuesday");
        break;

        case 3:
        printf("wednesday");
        break;

        case 4:
        printf("thursday");
        break;

        case 5:
        printf("friday");
        break;

        case 6:
        printf("saturday");
        break;

        case 7:
        printf("sunday");
        break;

        default:
        printf("not a valid number");
    }

    return 0;
}
```

```
enter a number from 1-7: 6
saturday

...Program finished with exit code 0
Press ENTER to exit console.
```

Ex. 2

Q.1)
Decision making: Choosing what code to execute based on a condition. It allows programs to make different choices.

Q.2)
if-else: checks one condition.
else-if ladder: checks multiple conditions sequentially.

Q.3)
Nested if-else: When one decision depends on another.
Else-if ladder: When checking several independent alternatives.

Q.4) Switch data types: only int, char, and other integer/enum types. Not float or double.

Q.5) Stops execution of switch. Without it, execution falls through to the next case.

Q.6) default executes when none of the cases match.

Q.7) max = (a > b) ? a : b;

Q.8) A

Q.9) switch requires an integral or enumeration type. Floating point values aren't allowed.

Q.10)
= : assignment
== : comparison